Medical Science Educator (2024) 34:691–694
https://doi.org/10.1007/s40670-024-02034-8

COMMENTARY

Assessing Risk of Bias Using ChatGPT-4 and Cochrane ROB2 Tool

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Accepted: 29 March 2024 / Published online: 5 April 2024
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Abstract
In the world of evidence-based medicine, systematic reviews have long been the gold standard. But they have had a problem—they take forever. That is where ChatGPT-4 and automation come in. They are like a breath of fresh air, speeding things up and making the process more reliable. ChatGPT-4 is like having a super-smart assistant who can quickly assess bias risk in research studies. It is a game-changer, especially in a field where getting the latest research quickly can mean life or death for patients. Sure, it is not perfect, and we still need humans to keep an eye on things and ensure everything's ethical. But the future looks bright. With ChatGPT-4 and automation, evidence-based medicine is on the fast track to success.

Keywords Artificial intelligence · Systematic review · Bias · Medical education

Introduction

Systematic reviews hold a pivotal position in medicine and healthcare, serving as the bedrock of evidence-based decision-making [1]. These comprehensive analyses systematically collate and assess existing research to provide invaluable insights into the effectiveness and safety of various medical treatments and interventions [1, 2]. As medical knowledge expands exponentially, the need for swift and efficient synthesis of emerging evidence has become more critical than ever before. Evidence synthesis, at its core, plays a vital role in bridging the gap between scientific research and clinical practice. It ensures that healthcare professionals, policymakers, and clinicians can access up-to-date and accurate information to make informed decisions about patient care [3]. However, traditional systematic review methods are often time-consuming and resource-intensive, leading to a lag in disseminating crucial findings. This delay can have significant consequences, potentially impacting patient outcomes and the quality of care provided.

In this era of rapid technological advancement, machine learning (ML) and artificial intelligence (AI) are emerging as promising tools to expedite the evidence-synthesis process [4]. These methods aim to reduce the substantial resources and time required to produce systematic reviews while maintaining the highest standards of rigor and quality. Among the key elements of a systematic review is the thorough assessment of included studies' risks of bias, a critical step in determining the reliability of the synthesized evidence. One of the most widely utilized tools for systematic reviews of randomized controlled studies is the Cochrane Collaboration's tool for assessing the risk of bias [5]. This tool employs a series of structured questions to evaluate various aspects of study design and conduct, helping reviewers identify potential sources of bias that could influence study outcomes. In this era of innovation, ChatGPT-4, a cutting-edge AI model, emerges as a valuable tool for researchers conducting systematic reviews. By harnessing AI capabilities, ChatGPT-4 swiftly and accurately addresses critical questions regarding bias assessment in research articles. Through automation, it streamlines the assessment process, improving efficiency and consistency.

The Impact of ChatGPT-4 and Automation on Risk of Bias Assessment in Medical Research

Integrating ChatGPT-4 and automation techniques into the systematic review process to assess bias risk heralds a groundbreaking paradigm shift in evidence synthesis within the medical field. This innovative approach offers numerous advantages, not only amplifying efficiency but also upholding the integrity of the systematic review process. Recent

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studies have highlighted the feasibility of combining human effort with machine automation to improve the productivity of living systematic reviews [6].

First, the infusion of ChatGPT-4 and automation ushers in remarkable efficiency gains. Conventional manual evaluations of bias risk can be both time-intensive and laborious, often leading to delays in evidence synthesis. With ChatGPT-4, we witness the rapid generation of critical risk of bias queries, significantly expediting the evaluation of many articles. This time-saving aspect is essential in healthcare decision-making, where timely access to the latest research can directly influence patient outcomes. Moreover, automation ensures a consistent and standardized approach to the risk of bias assessment. Human reviewers may inadvertently introduce variations or subjectivity into their evaluations. In contrast, ChatGPT-4 consistently adheres to predefined criteria, mitigating the risk of errors and ensuring high standardization across assessments. This consistency is pivotal in maintaining the integrity and reliability of systematic reviews.

Scalability emerges as another compelling advantage of the automated approach, especially when dealing with a substantial volume of articles. Systematic reviews frequently encompass the evaluation of numerous studies, and the ability to efficiently assess a significant body of literature is paramount [7].

ChatGPT-4's scalability empowers researchers to navigate complex reviews more quickly, making it an invaluable tool for comprehensive evidence synthesis. ChatGPT-4's usefulness is its ability to offer dependable bias assessments, drawing from a vast knowledge base. However, it is important to note that, like all AI systems, ChatGPT-4 could be more flawless. There may be occasional inaccuracies, highlighting the importance of human oversight and validation studies to ensure its reliability.

Importance and Application in Medical Education

In the realm of medical education, the integration of ChatGPT holds profound significance. Its application in this domain extends beyond mere novelty, offering transformative potential that could reshape the way medical students learn and engage with complex concepts, particularly in the context of systematic reviews.

ChatGPT's importance in medical education lies in its capacity to automate and streamline various facets of the learning process. By leveraging its advanced capabilities, educators can create immersive learning experiences, from generating virtual patient simulations to facilitating the creation of systematic reviews. This enhances the efficiency of educational endeavors and fosters a more profound understanding and retention of critical medical concepts among students. Furthermore, ChatGPT's utility extends to systematic reviews within medical education. Its ability to generate Boolean queries for literature searches and automate tasks such as screening, data extraction, and content analysis offers immense potential to expedite and streamline the systematic review process [8]. Automating tasks such as literature search, screening, data extraction, and content analysis allows ChatGPT to substantially diminish the time and effort invested in systematic reviews, widely regarded as the cornerstone of evidence in medicine [9]. By reducing the time and effort required for these endeavors, ChatGPT enables educators and students alike to conduct more comprehensive evidence synthesis, thereby fostering a deeper understanding of medical research methodologies and principles. Additionally, ChatGPT's contextual understanding and ability to provide logical justifications for answers can enhance the depth and accuracy of the risk of bias assessments. By contextualizing information and considering nuanced factors that contribute to bias in research studies, ChatGPT enables researchers to conduct more comprehensive and insightful evaluations, ultimately leading to more robust evidence synthesis.

Limitations and Mitigation Strategies

The effectiveness of ChatGPT-4 and automation in risk of bias assessment heavily relies on the quality and relevance of the input data. Inaccurate or biased input can lead to erroneous assessments and compromise the reliability of the entire process. One of the primary limitations is ChatGPT's inability to reason beyond the knowledge it has been trained on [10]. This is particularly problematic in systematic reviews, where interpreting new and complex data is paramount. Additionally, the generation of incorrect information by ChatGPT poses a risk to the integrity of systematic reviews, which rely on accurate data synthesis.

Moreover, ChatGPT-4 and automation may need help to grasp the nuanced contextual factors contributing to bias in research studies. Understanding the intricacies of study design, population characteristics, and intervention protocols requires human expertise that may only be partially replicable by AI systems. This limitation underscores the importance of human involvement in the risk of bias-assessment processes, particularly in areas where complex contextual understanding is essential.

Enhancing mitigation strategies is vital to addressing these limitations. This could involve advancing AI models to better comprehend context and nuances, alongside implementing stringent validation processes to verify the accuracy of ChatGPT-generated information. Moreover, establishing clear ethical guidelines and positioning AI as a supportive

tool rather than a substitute for human judgment could mitigate risks associated with its integration into systematic reviews.

Ethical and Legal Considerations

Ethical concerns surrounding the use of automation in risk of bias assessment stem from several angles. Legally, there are issues related to the unclear allocation of responsibility in cases of patient harm and potential breaches of patient privacy due to data collection [11]. The integrity of the physician–patient relationship may also be compromised, with overreliance on AI potentially eroding trust and compassion in healthcare [11]. Moreover, algorithmic bias presents a significant concern, as it can lead to skewed results and potentially harmful outcomes if not properly addressed [12]. Information ethics, including data bias and validity, are critical considerations, as biased training data can result in misleading outputs that may mislead patients and healthcare providers.

From a legal perspective, using ChatGPT in systematic reviews raises questions about copyright infringement and medico-legal complications. Ensuring transparency in AI-generated content is essential to avoid misleading healthcare professionals and patients [11]. Additionally, defining legal boundaries regarding the responsibility for AI-generated errors or misinformation is crucial to protecting users and determining liability. Establishing comprehensive ethical guidelines is essential to ensure the responsible utilization of ChatGPT in systematic reviews. These guidelines should prioritize accurate and reliable information exchange, safeguard patient privacy, and enable patients to make informed healthcare decisions [11]. Furthermore, before widespread adoption, thorough discussions and improvements regarding ChatGPT's accuracy, originality, academic integrity, and ethical considerations in medical research are imperative.

Conclusion

Integrating ChatGPT-4 and automation techniques into systematic reviews for risk of bias assessment heralds a promising era in medical evidence synthesis. Notably, ChatGPT-4 expedites the review process, alleviating the burden of manual, time-consuming evaluations and playing a pivotal role in healthcare decision-making, where timely access to research is paramount. Its scalability is invaluable in effectively managing large volumes of studies and streamlining the review process. While ChatGPT-4 has the potential to excel in risk of bias assessments, it is imperative to address ethical considerations, conduct validation studies, and perform cost–benefit analyses. Ensuring transparency and reliability in its utilization is paramount to its credibility.

In conclusion, the integration of ChatGPT-4 revolutionizes systematic reviews, enhancing their efficiency and quality. These advancements significantly affect healthcare policy, clinical decision-making, and patient outcomes. ChatGPT-4 empowers policymakers and clinicians to make well-informed decisions by providing timely access to credible research, ultimately improving patient outcomes. Despite challenges, the integration of ChatGPT-4 offers a promising future for evidence-based medicine, reshaping the landscape of medical research and healthcare delivery.

Acknowledgements I extend my heartfelt thanks to Maria Fernanda Ovalle for her unwavering support.

Availability of Data and Material N/A.

Code Availability N/A.

Declarations

Human and Animal Rights and Informed Consent The research does not involve human participants and/or animals.

Competing Interests The author declares no competing interests.

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